Kazi Tahsin Mahmood, Ph.D.

Assistant Teaching Professor
School of Engineering, Bowling Green State University

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SUMMARY

Assistant Teaching Professor of Mechanical Engineering and researcher working at the intersection of nonlinear dynamics, topological mechanics and acoustics, and quantum-inspired information science.

- Research focuses on developing theoretical, numerical, and experimental approaches for nonlinear elastic and acoustic systems, with emphasis on geometric phase, superposition of states, state-space dynamics, topological behavior, and wave control.
- Developed elastic-bit and phi-bit platforms for realizing classical analogues of quantum states and multi-state operations, with applications in sensing, metamaterials, and physical information processing.
- Dedicated to advancing student development through research mentorship, project-based learning, hands-on experimentation, and training in computational modeling, CAD, mechanics, and quantum information science.

ACADEMIC CREDENTIALS

Wayne State University **Detroit, Michigan, USA**
Ph.D. *in Mechanical Engineering* *Jan 2022 – Dec 2025*
Dissertation Title: *Classical Realization of Quantum and Topological States through Nonlinear Elastic Networks*
Committee: Dr. M. Arif Hasan (Chair); Dr. Sean Wu; Dr. Chin-An Tan; Dr. Mohammad Bukhari; Dr. Pierre Deymier

Wayne State University **Detroit, Michigan, USA**
M.Sc. *in Mechanical Engineering* *Jan 2022 – May 2025*
Thesis Title: *Nonlinear-Dynamics-Driven Quantum Analogues in Classical Granular Metamaterials*
Advisor: Dr. M. Arif Hasan

Bangladesh University of Engineering & Technology **Dhaka, Bangladesh**
B.Sc. (*Engg.*) *in Mechanical Engineering* *Feb 2016 – Mar 2021*
Thesis Title: *Electro-Thermo-Mechanical Response of a Thin Conductive Layer of Metal Matrix Composite Reinforced by Graphite Fibers*
Advisor: Dr. Shaikh Reaz Ahmed

RESEARCH AREAS

Topological Metamaterials	Quantum-Analogous Computing	Nonlinear Dynamics
Elastic & Acoustic Waveguides	Vibration & Acoustics	Solid Mechanics

PUBLICATIONS

1. **Kazi T. Mahmood** and M. Arif Hasan, "Geometric-phase diagnosis of mass defects through topological states in elastic granules," *Chaos, Solitons & Fractals*, vol. 212, pt. 2, p. 119024, 2026.
2. **Kazi T. Mahmood** and M. Arif Hasan, "Harnessing Nonlinear Dynamics for Time-Driven Berry Phase in Classical Systems," *Nonlinear Dynamics*, vol. 114, p. 689, May 2026.
3. **Kazi T. Mahmood**, M. Afridi Hasan, Abrar N. E. Faiaz, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Multi-Bit Quantum-Inspired Dynamics in Nonlinear Mechanical Oscillators," *Journal of Applied Mechanics*, vol. 93, no. 6, p. 061003, May 2026.
4. **Kazi T. Mahmood** and M. Arif Hasan, "Topological Vibration Analysis of Elastic Lattices via Bloch-Sphere Mapping," *Journal of Vibration and Acoustics*, vol. 148, no. 4, p. 041004, Feb. 2026.
5. **Kazi T. Mahmood**, Abrar N. E. Faiaz, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Experimental realization of logical elastic bits as qubit analogues in a nonlinear oscillator," *Scientific Reports*, vol. 16, Art. no. 3398, Dec. 2025.

6. **Kazi T. Mahmood** and M. Arif Hasan, "Topological insights from state manipulation in a classical elastic system," *AIP Advances*, vol. 15, no. 2, p. 025305, Feb. 2025.
7. **Kazi T. Mahmood** and M. Arif Hasan, "Experimental demonstration of classical analogous time-dependent superposition of states," *Scientific Reports*, vol. 12, no. 1, p. 22580, Dec. 2022.
8. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, and M. Arif Hasan, "Page-curve-like entropy dynamics in a classical elastic bit lattice," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 36, no. 4, p. 043127, Apr. 2026.
9. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Experimental Geometric State Control on the Bloch Sphere via Linear Mechanical Resonator," Accepted: *Journal of Vibration and Acoustics*, 2026.
10. **Kazi T. Mahmood** and M. Arif Hasan, "Phase-Sensitive Validation of Reduced-Order Nonlinear Mechanical Models with Dynamic Readout," Under review: *Journal of Computational and Nonlinear Dynamics*.
11. Abrar N. E. Faiaz, Akinsanmi S. Ige, **Kazi T. Mahmood**, Jake Balla, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Modeling Phi-Bits, Qubit Analogues, in Acoustic Waveguides," Under review: *Applied Acoustics*.
12. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, M. Arif Hasan, and Pierre A. Deymier, "Phase-Pattern Reversals and Topological Analogies in a Symmetry-Broken Coupled Resonator Network," Under review: *Journal of Vibration Engineering and Technology*.

CONFERENCE PRESENTATIONS

- "Classical analogous quantum coherent superposition through the Hertz-type nonlinearity of elastic granules," 184th *Meeting of the Acoustical Society of America*, Chicago, Illinois, USA, 2023 (**Technical Talk**).
- "Decoherence-Resistant Elastic Bits: A Novel Approach to Quantum Analogue Computing Using Coherent Superpositions of States," *International Conference on Smart Mobility and Vehicle Electrification*, Lawrence Technological University, Detroit, Michigan, USA, 2023 (**Technical Talk, Hybrid**).
- Graduate Research Symposium, Wayne State University, USA, 2024 (**Poster Presentation**).
- "Observation of Berry's Phase in Elastic Bits: Quantum Analogues in Classical Granular Systems," *APS March Meeting*, Minneapolis, USA, 2024 (**Technical Talk, Online**).
- "Exploration of the Quantum Coherent States Through Hertz-Type Classical Nonlinearity," *APS Eastern Great Lakes Spring Meeting*, Kettering University, Flint, Michigan, USA, 2024 (**Technical Talk**).
- "Quantum Concepts in Classical Realms: Berry Phases and Elastic Bits in Granular Systems," *APS Eastern Great Lakes Spring Meeting*, Kettering University, Flint, Michigan, USA, 2024 (**Poster Presentation**).
- "Elastic Bits and Berry Phases: Investigating Topological Phenomena in a Classical Granular Network," 186th *Meeting of the Acoustical Society of America/Acoustics Week in Canada*, Ottawa, Ontario, Canada, 2024 (**Technical Talk**).
- "Realization of Berry Phase in Classical Nonlinear Granular Network Using Quantum Analogue Theory," *APS Eastern Great Lakes Fall Meeting*, Marietta College, Marietta, Ohio, USA, 2024 (**Technical Talk**).
- "Bridging Classical Nonlinearity and Quantum Superposition," *International Mechanical Engineering Congress & Exposition*, Portland, Oregon, USA, 2024 (**Technical Talk**).
- "Quantum Realm in Classical Mechanics: A New Basis of Topological Computing," *International Mechanical Engineering Congress & Exposition*, Portland, Oregon, USA, 2024 (**Poster Presentation**).
- "The Notion of Berry Phase in Classical Analogues System," 187th *Meeting of the Acoustical Society of America*, 2024 (**Technical Talk, Online**).
- "Classical Granular System Topology: Berry Phase Formation in Elastic Bit," *International Electronic Conference on Applied Sciences*, 2024.
- NewFoS Research Symposium, University of Arizona, Tucson, Arizona, USA, January 2025 (**Poster Presentation**).
- "Elastic Bits in Granular Metamaterials: A Quantum-Inspired Framework for Robust Topological Computing," 188th *Meeting of the Acoustical Society of America*, New Orleans, Louisiana, USA, 2025 (**Technical Talk**).
- "Modeling Phi-bits in Acoustic Metamaterials for Quantum-Inspired Computation," 188th *Meeting of the Acoustical Society of America*, New Orleans, Louisiana, USA, 2025 (**Technical Talk**).
- "Geometric Phase and Topological Control of Elastic Lattices on the Bloch Sphere," *APS Global Physics Summit*, Denver, Colorado, USA, 2026 (**Technical Talk**).
- "Hilbert space framework for geometric phases and topological dynamics in mechanical lattices," 190th

Meeting of the Acoustical Society of America, Philadelphia, Pennsylvania, USA, 2026 (**Technical Talk**).

- "Exploring phi-bits in coupled acoustic waveguides: A classical platform for quantum-inspired information processing," 190th *Meeting of the Acoustical Society of America*, Philadelphia, Pennsylvania, USA, 2026 (**Technical Talk**).
- "State-Space Geometry and Geometric Phase in Static and Modulated Elastic Lattices," *APS Eastern Great Lakes Section (EGLS) & MIAAPT Joint Meeting*, Wayne State University, Detroit, Michigan, USA, 2026 (**Technical Talk**).

PROFESSIONAL EXPERIENCE

Assistant Teaching Professor **Bowling Green, Ohio, USA**
Department of Mechanical and Manufacturing Engineering
Bowling Green State University *Aug 2026 – Present*

- **Undergraduate Instruction:** Teach courses across three engineering areas:
  - *Design & Solid Modeling:* MME 1100 (Basic CAD), MME 1700 (Engineering Design and Skill), and ENGT 2100 (Solid Modeling).
  - *Core Mechanics:* MME 2400 (Statics) and MME 3350 (Machine Design).
  - *Manufacturing & Metrology:* MME 2500 (Metrology and GD&T) and ENGT 4250 (Manufacturing Design and Operations).
- **Curriculum Development:** Develop course syllabi, lectures, assignments, and projects aligned with undergraduate engineering learning outcomes.
- **Engineering Software & Analysis:** Integrate SolidWorks and AutoCAD into instruction to develop students' technical drawing and three-dimensional modeling skills; teach finite element analysis (FEA) for engineering analysis and design evaluation.
- **Experiential Learning:** Develop hands-on laboratory experiments, instrumentation activities, and design-for-manufacturing projects that connect engineering theory with physical components and assemblies.
- **Assessment & Student Support:** Develop and administer quizzes, examinations, and project assessments; support student learning through individualized feedback, regular office hours, and inclusive teaching practices.

Research Collaborator **Tucson, Arizona, USA**
New Frontiers of Sound (NewFoS)
The University of Arizona *Sep 2023 – Present*

NewFoS Project 1: *Topological Acoustic (TA) Quantum Analogies for Quantum Information Science (QIS)*

Elastic-Bit Feedback Control

- Fabricated and characterized a two-mass, three-spring mechanical resonator with independently controlled boundary excitation; integrated synchronized signal generation, vibration actuators, and scanning laser Doppler vibrometry to measure the amplitude and phase of the mass responses.
- Reconstructed symmetric and antisymmetric modal superpositions from frequency-domain velocity measurements and mapped their relative weights and phases onto the Bloch sphere. Applied a feedback procedure using a calibrated local Jacobian to update boundary-drive settings, with energy-based settling checks before each measurement.
- Demonstrated recovery of the prescribed Bloch state under persistent added-mass, 10% drive-amplitude, and 30° drive-phase disturbances; achieved geodesic errors below 0.01 rad after five, three, and five corrections, respectively, using the same nominal sensitivity calibration.

Real-Time Feedback Control of Acoustic Phi-Bits

- Pursue real-time, measurement-based feedback control of acoustic phi-bits in externally driven, coupled waveguides, with the goal of stabilizing phase-encoded states and understanding the nonlinear dynamics governing their response.
- Investigate inter-waveguide phase differences and relative amplitudes of nonlinear spectral modes as feedback observables, with drive-frequency and relative-phase adjustments as candidate control inputs. Examine how nonlinear phase sensitivity, phase jumps, and hysteresis affect state tracking and disturbance rejection to guide controller design.

Digital Twin for Disturbance-Robust State Preparation

- Develop a physics-based, steady-state digital twin of the two-mass elastic-bit platform to identify persistent mechanical and drive disturbances and predict excitation settings for prescribed Bloch states. Formulate validation using frequency-response measurements, laser Doppler vibrometry, modal reconstruction, and independent drive sweeps to assess model accuracy and disturbance-dependent sensitivity.
- Analyze parameter identifiability and investigate Kalman-type estimation using multiple probe excitations to distinguish mechanical disturbances from drive-channel gain and phase errors. Conduct preliminary simulations of damping and force-gain estimation from noisy frequency-response data, and examine how incorrect disturbance diagnosis affects prediction at previously unmeasured target states.
- Design inverse-drive sequences for quantum-analogous Bloch-sphere paths, including Pauli-X and Y rotations, as sequences of settled target states. Develop comparisons of nominal commands, measured-state feedback, twin-only prediction, and hybrid correction using geodesic error, corrections per target, measurement requirements, and total preparation time.

Postdoctoral Researcher **Detroit, Michigan, USA**

Acoustic Pseudospin and Metamaterials Laboratory
Wayne State University *Jan 2026 – Aug 2026*

- **Project 1: Phase-Sensitive Reduced-Order Modeling.** Developed and numerically validated a framework for nonlinear mechanical lattices coupled to dynamic readout channels. Used Galerkin projection, modal bases, and proper orthogonal decomposition (POD) to distinguish mechanical truncation error, readout prediction error, and distortion of normalized two-mode states. Evaluated modal amplitudes, relative phase, and state fidelity across forcing levels and independent test frequencies, examining channel damping, gain mismatch, nonlinear readout, and measurement noise.
- **Project 2: Nonlinear Acoustic Battery Thermometry.** Developed a research framework for temperature sensing with periodic elastic waveguides operating near a band edge, combining second-harmonic amplitude ratios with geometric-phase diagnostics. Formulated analytical and finite-element studies and a controlled experimental validation plan to distinguish temperature-induced stiffness changes from drive-level variation and coupling drift, with conventional ultrasonic time-of-flight measurements as a benchmark for potential internal hot-spot detection in lithium-ion batteries.
- **Project 3: Programmable Elastic-Bit Co-Processor.** Developed co-processor concepts for low-power audio and machine-health sensing, using programmable modal superpositions for band-pass filtering, spectral shaping, and Mel/DCT-like feature extraction before digitization. Formulated reconfiguration through drive frequency, phase, amplitude, and gate time, accounting for damping and packaging losses. Designed a validation plan for keyword spotting and vibration anomaly detection to compare transform fidelity, task accuracy, reconfiguration latency, and energy consumption against digital baselines.

Research Mentor **NSF Project**

New Frontiers of Sound (NewFoS) *Jan 2023 – Aug 2026*

- Mentored and supervised undergraduate researchers through the NSF New Frontiers of Sound (NewFoS) Research Experience and Mentoring (REM) program, guiding them in hypothesis formulation, experimental planning, data acquisition, and interpretation of results in nonlinear and topological acoustics.
- Trained students in the design and analysis of elastic waveguide systems, including mode identification, dispersion characterization, and phase–amplitude mapping, to investigate classical analogues of quantum information states.
- Facilitated literature review sessions and data presentation workshops to strengthen students' ability to connect experimental observations with theoretical models and communicate findings effectively.
- Oversaw independent research projects culminating in poster presentations at institutional and national conferences, fostering early-career engagement in interdisciplinary acoustics and quantum-inspired mechanics.

Graduate Research Assistant **Detroit, Michigan, USA**

Acoustic Pseudospin and Metamaterials Laboratory
Wayne State University *May 2024 – Dec 2025*

- **Project 1:** Developed the **Elastic Bit**, a classical analog of a quantum bit, in granular-bead elastic media using linearized and nonlinear models. Constructed analytical and experimental nonlinear granular-

network models with **asymptotic perturbation techniques**, demonstrating time-dependent superposition responses for potential data transmission (NSF Grant 2204382).
- **Project 2:** Developed a theoretical framework for manipulating elastic-bit superposition states and generating **Berry phase** in linearized and nonlinear elastic networks, demonstrating quantum-classical correlations in time-dependent and time-independent systems (NSF Grant 2242925).
- **Project 3:** Developed a computational framework to investigate the scalability of **Logical Elastic Bits** in a nonlinear mass-conical-spring oscillator. Modeled and analyzed the system's harmonic response to construct time-dependent elastic-bit states and examine their extension to multi-bit, higher-dimensional state spaces.
- **Project 4:** Investigated symmetry breaking in granular networks using a **second-order molecular dynamics** algorithm, revealing internal parameter variations that drive topological transitions across the **Brillouin zone**. Studied multi-mass unit cells to explore the corresponding **Hilbert space**.

Graduate Teaching Assistant **Detroit, Michigan, USA**
Department of Mechanical Engineering
Wayne State University *Jan 2022 – May 2024*

- Served as **Course Instructor** for **ME 3400: Dynamics I** and **ME 5000: Engineering Analysis**, delivering lectures, designing problem sets, and evaluating student performance for both undergraduate and graduate cohorts.
- Led laboratory sessions for **ME 7650: Topological Acoustics**, guiding students through experimental procedures, data analysis, and the interpretation of topological acoustic phenomena.
- Provided instructional support for multiple laboratory and recitation sections, including grading, feedback, and individualized student mentoring.
- Assisted in exam administration and proctoring, ensuring academic integrity and smooth execution of assessments.

Engineer Trainee (Intern) **Chattogram, Bangladesh**
BSRM Steel Limited *Jan 2020 – Feb 2020*

- Gained hands-on experience in industrial steel-rod manufacturing, including billet reheating, multi-pass hot rolling, controlled water quenching, and atmospheric cooling for TMT reinforcement bars meeting ASTM and BSTI standards.
- Assisted in the **mechanical maintenance** of re-rolling mill equipment, including rolling stands, drive shafts, guide assemblies, and cooling-bed conveyors. Performed routine inspections, alignment, lubrication scheduling, and fault diagnostics to reduce unplanned downtime and support production safety.

Undergraduate Research Assistant **Dhaka, Bangladesh**
Department of Mechanical Engineering
Bangladesh University of Engineering & Technology *Nov 2019 – Feb 2021*

- Studied the strength of metal matrix composites (MMCs) subjected to electrical current through thin composite layers. Analyzed current-induced thermal stress and the effects of MMC mesh geometry.
- Analyzed stress tolerance limits and failure characteristics due to thermal and mechanical stresses in the material using a **tensile** and **compression** testing machine.
- Developed a **heat pipe** exchanger model to increase heat transfer efficiency in a compact thermal management system for a high-performance induction motor.

TECHNICAL SKILLS

- **Fabrication & Experimental:** Granular bead networks, mass-conical spring models, coupled linear waveguides, heat pipe exchanger, vibration testing design, topological mass-sensing prototype design.
- **Programming & Simulation:** C/C++, Python, MATLAB, LabVIEW, COMSOL, ANSYS FEA.
- **Mathematical Modeling:** Finite difference and finite element formulation, multi-order asymptotic perturbation modeling, Fourier harmonics, nonlinear dynamics modeling, structural dynamics.
- **CAD & Design:** SolidWorks, AutoCAD, CATIA, Creo, GD&T.
- **Instrumentation & Hardware:** Laser Doppler Vibrometer (Polytec), Oscilloscope (Tektronix), Function Generator (B&K Precision), Power Amplifier (Piezodrive), Vibration Shakers/Exciters, Transducers,

Synchronized Sensor Integration.
- **Standards & Methods:** Experimental data acquisition, noise isolation, preventive maintenance, diagnostics, hardware characterization, root-cause troubleshooting.
- **Productivity & Documentation:** LaTeX, Microsoft Office.

PROFESSIONAL SERVICES

Reviewer

- Scientific Reports, Nature Portfolio — Jan 2024 – Present
- Journal of Vibration & Acoustics, ASME — Jan 2025 – Present
- IDETC-CIE, ASME — 2026
- Discover Quantum Science, Springer Nature — Apr 2026 – Present

AWARDS

- Graduate Student Professional Travel Grant, Wayne State University, 2024.

PROFESSIONAL MEMBERSHIPS

- Member, American Society of Mechanical Engineers (ASME) — Jun 2024 – Dec 2025
- Member, American Physical Society (APS) — Jul 2023 – Dec 2025
- Member, Acoustical Society of America (ASA) — Jan 2023 – Present
- Executive Member, BUET Robotics Society — Feb 2016 – Jul 2020

REFERENCES

M. Arif Hasan, Ph.D.
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Department of Mechanical Engineering
Wayne State University
NSF CMMI Panel Fellow
Editorial Board Member at Scientific Reports (Nature)
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Pierre A. Deymier, Ph.D.
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